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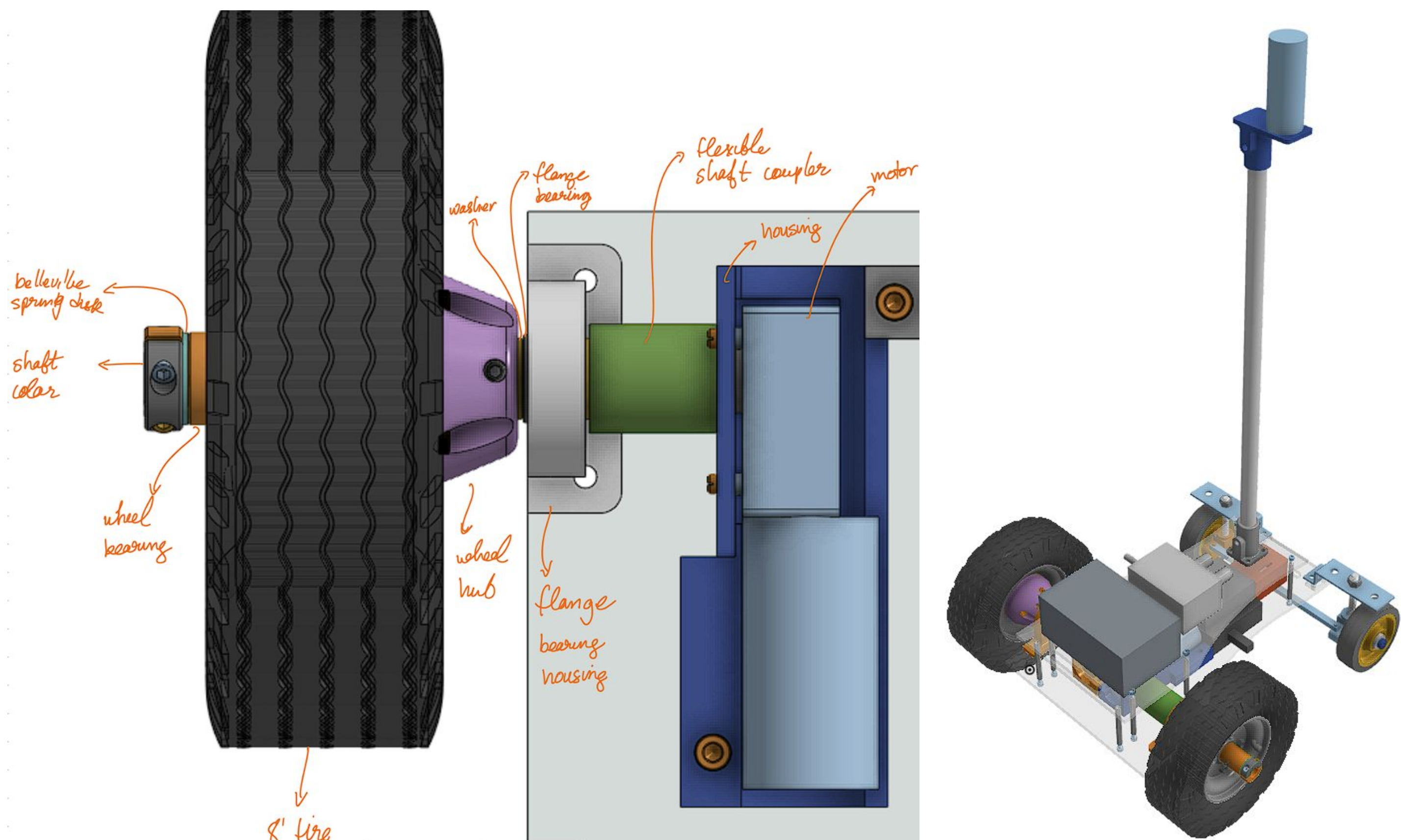
SnoScout: A Mechatronics Solution to Remotely Locating Buried Avalanche Victims

Background



Every year, over 150 people die in avalanches. Effective avalanche rescue depends on quickly locating buried victims. Buried victims begin asphyxiating within the first 15 minutes, yet rescuers currently search for 457kHz beacons worn by the victims using with a manual grid search over a large area, a process that can take hours in large avalanches. The SnoScout is a solution to this: A robot that effectively traverses avalanche terrain and rapidly locates beacons. The SnoScout will rapidly speed up searches by giving rescuers an aerial drone that can traverse the difficult terrain with sensors to quickly locate buried victims. In order to fit within time and budget constraints we made a proof of concept on a two wheel drive ground based rover. The rover uses a compact active receiving antenna, a Software Defined Radio (SDR) receiver, and a Raspberry Pi 4b to handle the beacon detection signal processing. It also has a GPS module and an ESP32 for control and localization. This data is fused into geospatial points which the robot uses to estimate beacon positions.

Hardware Architecture and CAD



Drive System: Our rover has a two-wheel direct drive system. It was chosen because it removes compliance and backlash from the drive train, which gives us a more reliable low-speed transmission while keeping the drive compact. Two 70RPM, high torque planetary gear motor power the transmission of the rover. The rated torque of 30kg*cm ensures the rover is able to climb mountains graded upto 25 degrees. The right angle configuration of the motors also allows for a compact drive.

Computers and Controllers: A Raspberry Pi 4 is dedicated to high-level signal processing, while an, ESP32 handles drive commands and communication.

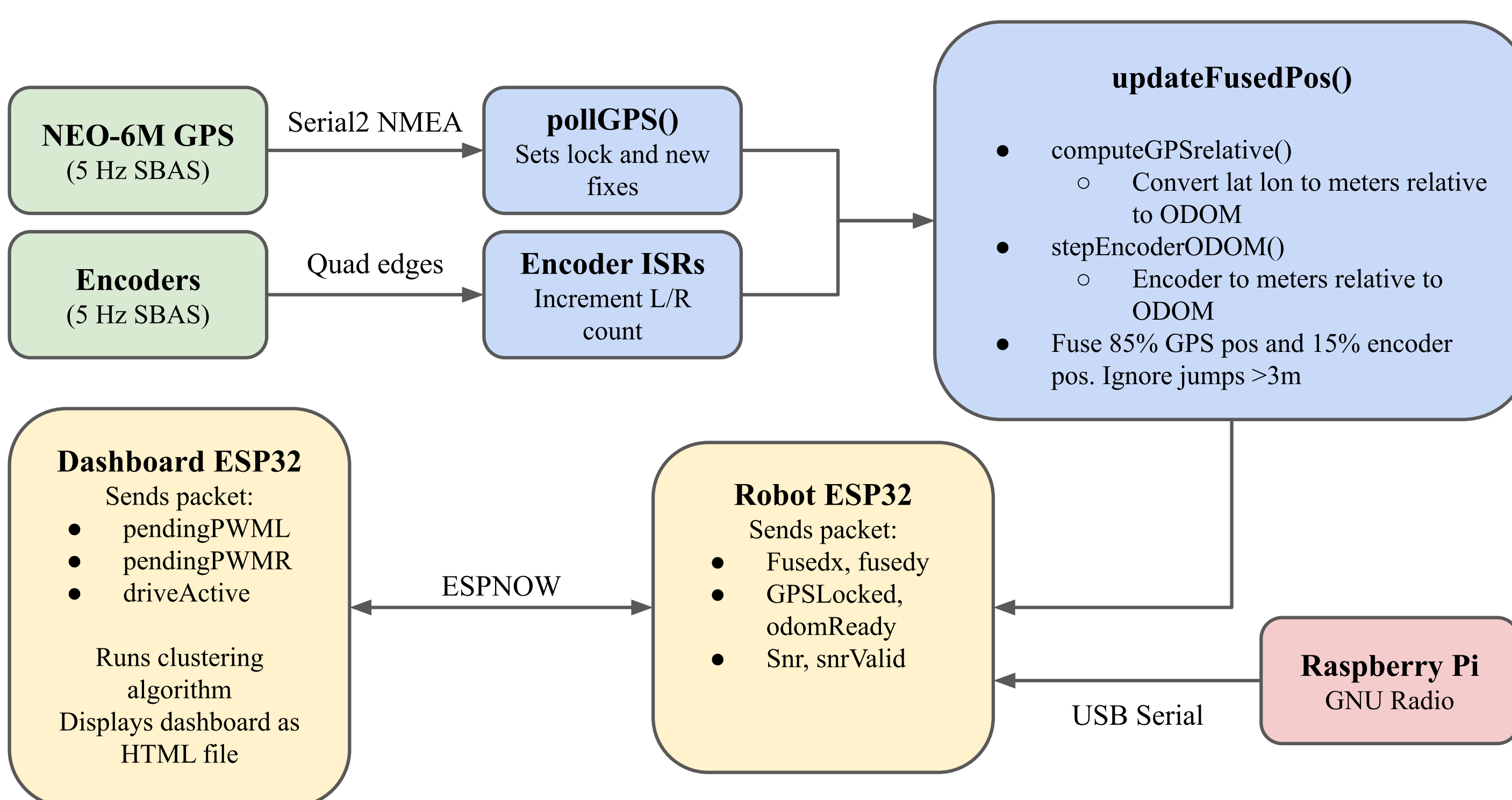
SDR Receiver: SDRPlay RSP1B is utilized for capturing high-fidelity IQ (In-Phase and Quadrature) data at the 457kHz beacon frequency.

Antenna: A PA0RDT Mini-Whip antenna is mounted for signal acquisition.

GPS: A NEO 6M GPS Module provides location tracking with 5 Hz, SBAS-corrected precision

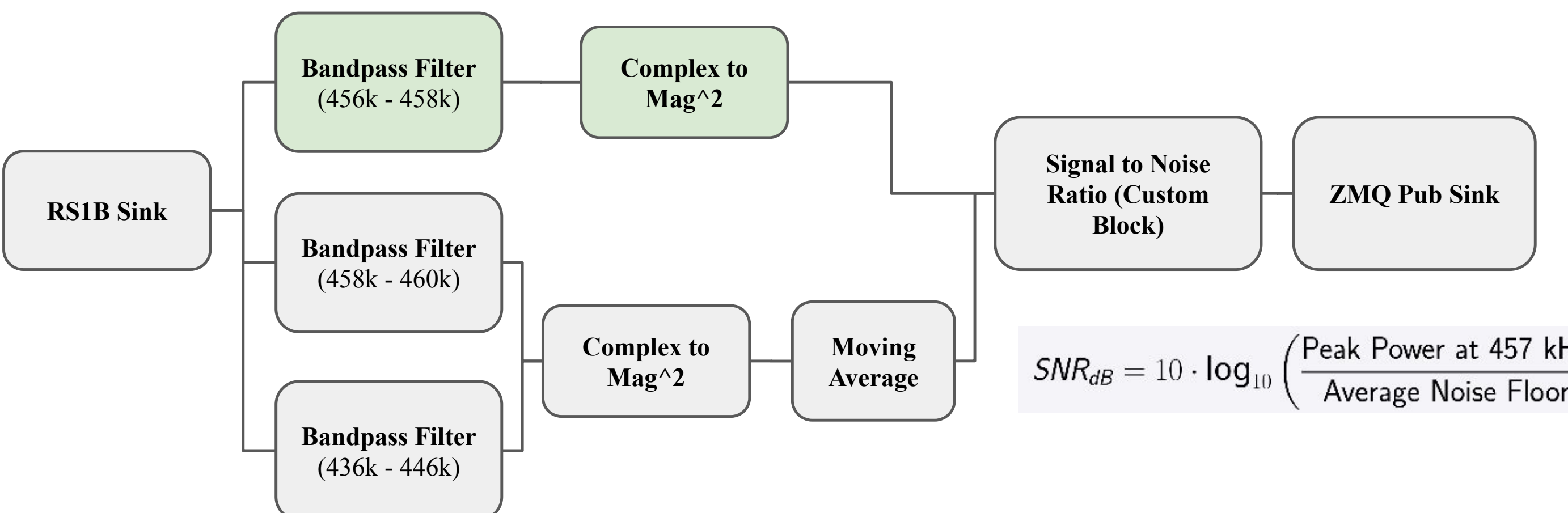
Chassis and Mounts: The frame consists of two laser cut ¼ inch acrylic layers separated by standoffs, with all electronics secured in custom 3D printed ABS mounts.

Data Pipeline and Signal Processing



System Diagram

The robot fuses position estimates from its wheel and encoders and GPS using a weighted blend. A Raspberry Pi running GNU Radio extracts the beacon signal and streams SNR readings to the robot ESP32. The robot ESP32 broadcasts fused position and SNR data to a Dashboard ESP32 which runs DBSCAN and presents a live dashboard while simultaneously sending motor commands back to the robot.



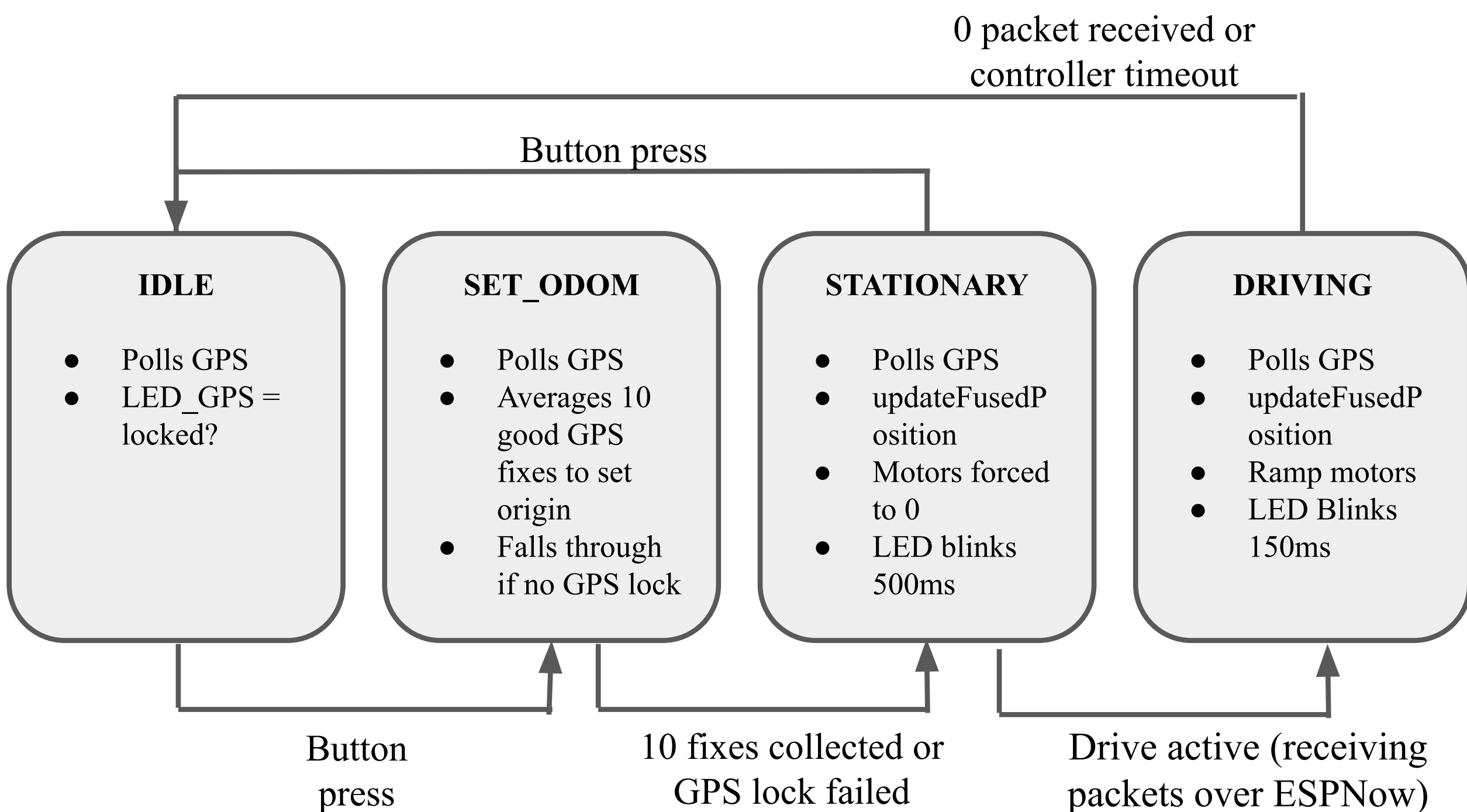
Signal Processing

The rover's Raspberry Pi uses a custom GNU flowgraph to process radio signals in real time:

Acquisition: continuous capture of the 457kHz frequency by the MiniWhip Antenna

Filtering: Average of two noise reference bands 10kHz from signal to calculate SNR (Signal Noise Ratio)

State Transition Diagram



Manufacturing

Open Chassis

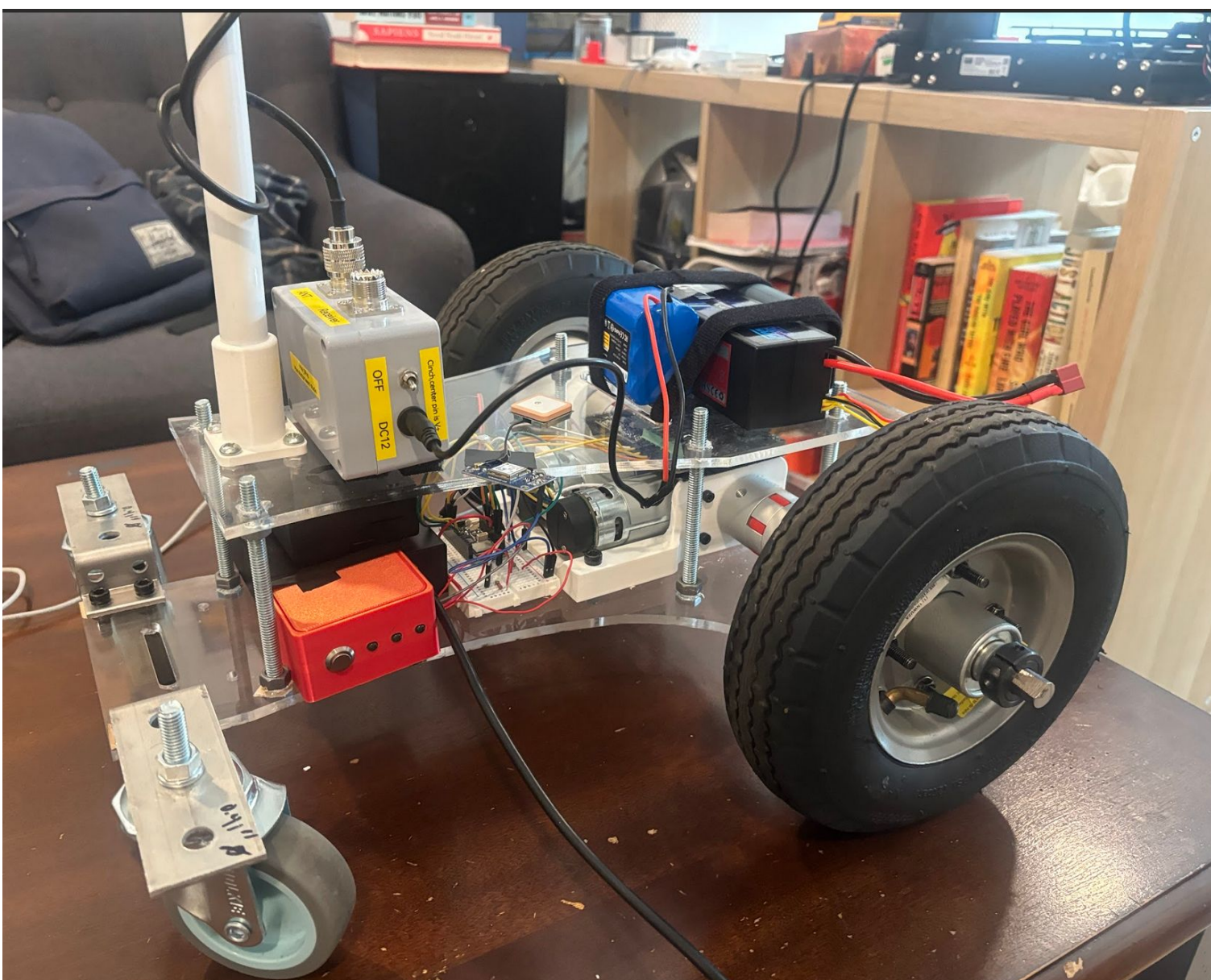
Our Chassis is made of two laser cut layers of ¼ extruded acrylic separated by standoffs. This allows for easy access, cheap manufacturing, and easy assembly for rapid prototyping.

Mounts

Every mount is made of 3D printed from ABS. The electronics use velcro harnesses for the ease of assembly while troubleshooting.

Transmission

We machined a flat into the transmission shaft to create a d-shaped shaft and taps for set screws into cheap flex shaft couplers. We also designed a 3D printed custom wheel hub with heat set inserts for set screws. These features prevent slipping and significantly reduce costs by using cheap off the shelf components.



Clustering Algorithm

Beacon locations are found using a set of geospatial points the robot collects using a two stage process. First, only points whose signal to noise ratio (SNR) exceed the 70th percentile are considered. This allows us to pick up weak signals while discarding any relatively weaker false detections. These filtered points are then clustered using DBSCAN (Density-Based Spatial Clustering of Applications with Noise) which groups nearby measurements while disregarding outliers. The final beacon position is determined using a weighted centroid of the cluster based on SNR strength.



The current DBSCAN algorithm assumes linear distance while in reality the signal is a function of distance squared. The beacon also only transmits once per second creating a data acquisition bottleneck.

Conclusions

The SnoScout successfully demonstrates a proof of concept for the rapid localization of avalanche victims using a robotic platform. Our system integrates Software Defined Radio (SDR) and GNU Radio to process 457kHz beacon signals in real-time. By fusing GPS and encoder data, the rover achieves reliable localization. The DBSCAN-based clustering algorithm effectively identifies beacon positions by prioritizing high-SNR data and filtering out environmental noise. This project validates that a mechatronic solution can significantly reduce search times, providing a foundation for future aerial drone integration to save lives in alpine terrain. We hope to continue working on the prototype to make it a functional product.

Future Work

- I. Our primary goal for the future of SnoScout is to package this system into a compact aerial drone which search and rescuers can use to navigate search and rescue missions in alpine terrain. That would turn this proof of concept into a product that could potentially save hundreds of lives.
- II. Another goal is to fine tune our search methods. We would replace our single multi directional antenna with three orthogonal antennas. This would improve our range while also letting us switch from a clustering algorithm to a gradient descent algorithm, massively cutting search time and fixing our signal bottleneck.
- III. A stretch goal we have is to research detection methods other than beacons such as phone detection, opening the use cases up to any rescue operation.